Supporting Information

SIG-Net: Similarity-Informed Local-Competence Mixtures of Cross-Fitted Experts for Molecular Property Prediction under Leakage-Audited Out-of-Distribution Splits

S1. Pre-specified analysis plan

The analysis plan below was written after the final SIG-Net run was launched and before any of its test-set results were inspected. The SIG-Net design was fixed using validation scores on three development folds.

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# Pre-specified analysis plan for SIG-Net v3

Written 2026-10-17 ~23:35 local time, after the v3 RunPod run was launched and **before any v3 tes
The v3 design (experts, cross-fitting K=5, grid for k/tau/gamma) was fixed using validation scores
folds (FreeSolv bm_scaffold s42, ESOL bm_scaffold s42, ESOL butina_cluster s42); no test scores we

## Data
- 4 datasets x 4 split protocols x 5 seeds = 80 folds; identical splits/fold preparation for v2 an
- v2 results (‘results/folds/‘) and v3 results (‘results/folds_v3/‘) are merged per fold.

## Primary endpoint
- SIG-Net v3 (Proposed) vs. the strongest non-v3 comparator, defined as the non-v3 model with the
  across the 16 settings. Test: two-sided Wilcoxon signed-rank test on per-fold relative improveme
  Also reported: number of settings where v3 has the better 5-seed mean.

## Secondary endpoints
1. v3 vs. "Global stack (same cross-fitted experts)" -- isolates the contribution of local compete
2. v3 vs. each bagged expert and each v2-run baseline (same Wilcoxon procedure, 80 folds).
3. Average rank over all models (Friedman test, Nemenyi CD).
4. Error stratified by nearest-neighbour Tanimoto similarity (OOD folds, regression datasets).

## Reporting rules
- All 16 settings reported, including those where v3 is not best.
- No post-hoc removal of datasets, splits, seeds or baselines.
- FreeSolv generic_scaffold has a single distinct test set across seeds; it is reported but flagge
- If v3 does not beat the strongest comparator on the primary endpoint, this is stated as the resu

## Addendum (2026-10-18 ~00:55, written before any BBBP or ChemBERTa-2 result was viewed)
Following external review of the draft, two extensions were added. The SIG-Net design (experts, K,
1. **BBBP** (MoleculeNet blood-brain barrier penetration, classification) is added as a fifth data
   with the identical Phase 1 code (4 protocols x 5 seeds). All models, including SIG-Net, are run
2. **Pretrained foundation-model baselines:** frozen ChemBERTa-2 (DeepChem/ChemBERTa-77M-MTR) mean
   (a) a ridge/logistic head and (b) an XGBoost head, both selected on the validation fold, on all
   ChemBERTa-2 is a comparison model only; it is not added as a SIG-Net expert (that would change
Reporting: the main text reports all 5 datasets (100 folds). The pre-specified 80-fold primary end
datasets, original comparator set) is reported unchanged alongside it. No other analysis changes.

## Addendum 2 -- SIG-Net v4 study (2026-10-18 ~09:40, before any confirmatory result was computed)
**Motivation:** reviewer request to simplify or justify the novelty exponent and to strengthen the
**Splits:** fresh seeds generated with the identical splitting code. Development seed **11** (all
for design decisions, including its test folds. Confirmatory seeds **1, 2, 3** (60 folds) are not
is frozen. FreeSolv generic_scaffold yields the same test set for every seed, so it is **excluded
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endpoint** (reported separately).
Experts: 9 cross-fitted experts (the 7 of v3 + ChemBERTa-2/XGBoost + XGBoost/ECFP4); all OOF/v
Candidate routers (development): v4 (local error-covariance routing with novelty shrinkage tow
covariance, composite ECFP/MACCS/physchem kernel, novelty-damped local residual correction), on 7
global stacks; and v4 ablations.
Design-selection rule (fixed now, before the remaining development folds arrived): the v4 vari
per-fold relative improvement over the 7-expert global stack across the 20 seed-11 folds is chosen
Confirmatory primary endpoint: chosen v4 vs the 7-expert global stack over the 56 confirmatory
generic), two-sided Wilcoxon signed-rank test on per-fold relative improvement. **Secondary:** v4
9-expert global stack; v4 ablations (no residual correction, no composite kernel, no novelty shrin
Reporting: whatever the confirmatory result is, it is reported; the v3 paper results are uncha

Design freeze (2026-10-18 09:54, before any confirmatory result was computed)
Applying the rule above to the 20 development folds (file results/dev_v4_seed11_FROZEN.csv): **v4
highest mean improvement over the 7-expert global stack (+0.68%, median +0.51%, 14/20 wins) and is
9 experts; grid k in {10,25,50}, a in {0.5,0.75,1.0}, gamma in {0,0.5,1,2,4,global}, nu in {0,0.25
gamma_d in {1,2}, rho = 0.05; all selected per fold on the validation fold. Ablations are those li

S2. Full benchmark

Table S1: Mean test metric over five seeds for every model (RMSE for FreeSolv/ESOL/Lipophilicity, ROC-AUC for BACE/BBBP). R: random, BM: Bemis–Murcko scaffold, G: generic skeleton, Bu: Butina cluster.

Model	Free-R	Free-BM	Free-G	Free-Bu	ESOL-R	ESOL-BM	ESOL-G	ESOL-Bu	Lipo-R	Lipo-BM	Lipo-G	Lipo-Bu	BACE-R	BACE-BM	BACE-G	BACE-Bu	BBBP-R	BBBP-BM	BBBP-G
Anchor (Multi-View Stack) [5-fold bagged]	0.964	1.870	1.430	1.083	0.545	0.767	0.753	0.577	0.594	0.687	0.688	0.791	0.901	0.855	0.798	0.763	0.931	0.937	0.950
Anchor-Only (Multi-View Stack)	0.939	1.870	1.430	1.083	0.545	0.767	0.753	0.577	0.595	0.687	0.688	0.791	0.902	0.855	0.798	0.763	0.931	0.937	0.950
AttentiveFP	0.983	2.030	1.363	1.097	0.570	0.849	0.807	0.602	0.545	0.614	0.634	0.724	0.872	0.836	0.806	0.704	0.892	0.898	0.923
AttentiveFP [5-fold bagged]	0.898	1.910	1.348	1.167	0.585	0.810	0.771	0.614	0.528	0.584	0.605	0.687	0.881	0.856	0.781	0.772	0.901	0.901	0.932
ChemBERTa-2 (77M-MTR) + XGBoost	2.009	3.026	2.569	1.786	0.847	1.128	0.994	0.839	0.743	0.873	0.860	0.964	0.868	0.838	0.819	0.800	0.907	0.914	0.936
ChemBERTa-2 (77M-MTR) + linear	1.386	2.309	1.809	1.397	0.793	1.038	1.289	0.839	0.768	0.892	0.842	0.976	0.885	0.840	0.799	0.730	0.876	0.895	0.932
D-MPNN (Chemprop)	1.111	1.823	1.337	1.219	0.652	0.861	0.890	0.641	0.527	0.646	0.636	0.717	0.887	0.840	0.731	0.745	0.906	0.929	0.912
D-MPNN + Deep CORAL	0.871	1.714	1.526	1.079	0.556	0.775	0.743	0.567	0.547	0.660	0.671	0.766	0.879	0.832	0.807	0.780	0.925	0.914	0.931
D-MPNN + Deep CORAL [5-fold bagged]	0.878	1.821	1.558	1.135	0.533	0.756	0.728	0.552	0.541	0.651	0.645	0.739	0.881	0.843	0.812	0.795	0.926	0.915	0.935
D-MPNN + RDKit-2D	0.871	1.723	1.525	1.089	0.557	0.777	0.736	0.568	0.544	0.665	0.670	0.764	0.879	0.830	0.812	0.781	0.926	0.916	0.927
D-MPNN + RDKit-2D Ensemble (3x)	0.873	1.759	1.537	1.107	0.536	0.773	0.748	0.560	0.540	0.653	0.660	0.747	0.879	0.837	0.813	0.782	0.925	0.921	0.927
D-MPNN + RDKit-2D [5-fold bagged]	0.870	1.795	1.556	1.115	0.536	0.762	0.730	0.565	0.542	0.648	0.649	0.744	0.883	0.840	0.816	0.782	0.926	0.914	0.934
Dual-Stream Gated Residual (v1 design)	0.980	1.792	1.432	1.114	0.542	0.799	0.760	0.566	0.571	0.680	0.665	0.791	0.900	0.855	0.799	0.758	0.932	0.939	0.944
GATv2	1.234	1.976	1.395	1.346	0.567	0.861	0.803	0.624	0.574	0.628	0.662	0.763	0.853	0.823	0.701	0.681	0.892	0.902	0.906
GCN	1.011	2.160	1.279	1.237	0.583	0.853	0.742	0.607	0.616	0.683	0.706	0.788	0.813	0.776	0.689	0.714	0.926	0.915	0.923
GINE	0.996	2.014	1.335	1.216	0.560	0.774	0.756	0.579	0.556	0.628	0.638	0.714	0.764	0.794	0.703	0.615	0.912	0.918	0.924
GINE [5-fold bagged]	0.978	1.996	1.381	1.238	0.563	0.819	0.740	0.593	0.557	0.610	0.632	0.722	0.862	0.831	0.716	0.720	0.916	0.921	0.938
Global stack (same cross-fitted experts)	0.822	1.719	1.465	1.081	0.516	0.763	0.696	0.544	0.506	0.576	0.590	0.670	0.901	0.863	0.810	0.767	0.935	0.939	0.952
HighwayMLP (ECFP4 + RDKit-2D)	0.971	1.812	1.565	1.064	0.561	0.821	0.771	0.609	0.575	0.655	0.679	0.745	0.888	0.852	0.782	0.767	0.914	0.917	0.932
RF (ECFP4-2048)	2.309	3.765	2.818	2.261	1.241	1.654	1.639	1.295	0.886	0.982	0.930	1.032	0.893	0.844	0.783	0.780	0.918	0.904	0.934
RF (MACCS-167)	1.141	1.905	1.690	1.389	1.058	1.334	1.359	1.002	0.800	0.918	0.896	0.958	0.880	0.830	0.772	0.734	0.926	0.930	0.943
RF (RDKit-2D)	1.148	2.349	1.878	1.324	0.623	0.835	0.832	0.645	0.668	0.765	0.771	0.879	0.880	0.846	0.790	0.793	0.925	0.932	0.948
RF (RDKit-2D) [5-fold bagged]	1.283	2.452	1.842	1.379	0.641	0.840	0.816	0.653	0.675	0.767	0.772	0.880	0.883	0.847	0.794	0.790	0.927	0.932	0.949
SIG-Net v2 (Proposed)	0.868	1.753	1.415	1.074	0.515	0.751	0.734	0.549	0.530	0.636	0.633	0.730	0.901	0.853	0.818	0.765	0.933	0.940	0.944
SIG-Net v2 w/o ensemble	0.878	1.743	1.397	1.065	0.523	0.746	0.722	0.554	0.530	0.641	0.635	0.733	0.902	0.854	0.817	0.765	0.934	0.938	0.946
SIG-Net v3 (Proposed)	0.800	1.720	1.405	1.038	0.511	0.754	0.704	0.544	0.505	0.580	0.589	0.668	0.900	0.864	0.815	0.790	0.940	0.940	0.954
SIG-Net v3 w/o novelty shrinkage	0.797	1.687	1.415	1.038	0.512	0.741	0.703	0.544	0.509	0.592	0.596	0.675	0.901	0.863	0.818	0.790	0.939	0.940	0.954
Stacked (Ensemble + Anchor, global w)	0.875	1.783	1.427	1.094	0.515	0.752	0.734	0.550	0.533	0.636	0.634	0.730	0.899	0.848	0.816	0.760	0.934	0.938	0.941
Stacked (XGBoost + D-MPNN)	0.884	1.880	1.474	1.057	0.532	0.751	0.717	0.563	0.531	0.637	0.636	0.735	0.892	0.841	0.807	0.762	0.931	0.935	0.936
XGBoost (ECFP4 + RDKit-2D)	1.066	2.464	1.773	1.157	0.597	0.788	0.787	0.594	0.597	0.693	0.698	0.809	0.889	0.849	0.789	0.759	0.926	0.933	0.949
XGBoost (ECFP4 + RDKit-2D) [5-fold bagged]	1.152	2.309	1.755	1.195	0.592	0.765	0.763	0.605	0.607	0.694	0.701	0.808	0.891	0.849	0.789	0.773	0.926	0.935	0.950
XGBoost (ECFP4-2048)	1.697	3.229	2.437	1.771	1.126	1.530	1.464	1.210	0.794	0.874	0.866	0.978	0.899	0.844	0.779	0.705	0.912	0.896	0.919
XGBoost (RDKit-2D)	0.998	2.251	1.585	1.099	0.596	0.792	0.787	0.604	0.609	0.700	0.715	0.823	0.882	0.843	0.777	0.742	0.924	0.928	0.949

S3. Full rankings

Table S2: Mean rank over 20 settings for all models except internal SIG-Net ablations.

#	Model	Mean rank	OOD mean rank	First	Top-3
1	SIG-Net	2.60	2.87	12	14
2	Two-expert novelty gate	5.20	5.80	1	7
3	Two-expert gate (single D-MPNN)	5.50	5.93	3	6
4	Stack: D-MPNN ens. + anchor	7.50	8.20	0	3
5	Stack: XGBoost + D-MPNN	9.65	10.27	0	3
6	Multi-view anchor (5-fold bagged)	11.38	11.73	0	2
7	Multi-view anchor	11.62	12.07	0	1
8	D-MPNN + Deep CORAL [5-fold bagged]	11.65	12.00	0	1
9	D-MPNN + RDKit-2D [5-fold bagged]	11.90	12.73	0	1
10	Dual-stream gated residual	12.25	12.87	0	1
11	D-MPNN + RDKit-2D Ensemble (3x)	13.35	13.53	0	0
12	D-MPNN + RDKit-2D	14.20	14.27	0	1
13	D-MPNN + Deep CORAL	14.30	14.33	1	1
14	AttentiveFP [5-fold bagged]	14.85	14.33	0	5
15	HighwayMLP (ECFP4 + RDKit-2D)	16.40	16.33	0	1
16	GINE [5-fold bagged]	16.55	15.67	0	2
17	XGBoost (ECFP4 + RDKit-2D) [5-fold bagged]	16.70	16.13	0	1
18	XGBoost (ECFP4 + RDKit-2D)	18.05	18.00	0	0
19	GINE	18.30	17.60	0	2
20	D-MPNN (Chemprop)	18.45	18.53	0	2
21	AttentiveFP	18.50	17.80	0	0
22	XGBoost (RDKit-2D)	19.10	18.67	0	0
23	RF (RDKit-2D) [5-fold bagged]	19.70	19.40	0	0
24	RF (RDKit-2D)	19.90	19.07	0	1
25	GCN	21.10	21.13	1	1
26	GATv2	22.30	21.87	0	0
27	RF (MACCS-167)	23.10	22.93	0	1
28	ChemBERTa-2 (77M-MTR) + XGBoost	23.50	22.13	2	2
29	ChemBERTa-2 (77M-MTR) + linear	25.55	25.47	0	0

S4. All paired comparisons (100 folds)

Table S3: SIG-Net vs. every other model. Relative improvement per fold (%), two-sided Wilcoxon signed-rank test.

Comparator	Folds won	Settings won	Mean	Median	p
SIG-Net w/o novelty shrinkage	27/100	9/20	+0.05	+0.00	$p = 0.166$
Global stack (same experts)	60/100	15/20	+0.80	+0.14	$p = 0.0035$
Two-expert gate (single D-MPNN)	68/100	14/20	+2.63	+1.58	$p = 1.6 \times 10^{-5}$
Two-expert novelty gate	69/100	15/20	+2.67	+1.47	$p = 8.2 \times 10^{-6}$
Stack: D-MPNN ens. + anchor	73/100	19/20	+3.18	+1.89	$p = 2.0 \times 10^{-7}$
Stack: XGBoost + D-MPNN	80/100	19/20	+3.85	+2.75	$p = 1.3 \times 10^{-9}$

D-MPNN + Deep CORAL [5-fold bagged]	86/100	19/20	+4.57	+3.67	$p = 2.0 \times 10^{-13}$
D-MPNN + RDKit-2D [5-fold bagged]	88/100	19/20	+4.72	+4.38	$p = 1.6 \times 10^{-13}$
D-MPNN + RDKit-2D Ensemble (3x)	83/100	20/20	+4.87	+4.50	$p = 1.7 \times 10^{-11}$
D-MPNN + Deep CORAL	82/100	19/20	+5.18	+4.64	$p = 3.8 \times 10^{-11}$
D-MPNN + RDKit-2D	84/100	19/20	+5.23	+4.72	$p = 2.4 \times 10^{-11}$
AttentiveFP [5-fold bagged]	87/100	19/20	+5.31	+3.87	$p = 9.2 \times 10^{-12}$
Dual-stream gated residual	80/100	19/20	+5.45	+2.56	$p = 2.4 \times 10^{-11}$
Multi-view anchor	76/100	18/20	+5.70	+2.48	$p = 6.2 \times 10^{-10}$
Multi-view anchor (5-fold bagged)	76/100	18/20	+5.83	+2.59	$p = 3.3 \times 10^{-10}$
HighwayMLP (ECFP4 + RDKit-2D)	87/100	20/20	+6.95	+5.93	$p = 3.0 \times 10^{-12}$
AttentiveFP	87/100	19/20	+7.41	+6.32	$p = 1.7 \times 10^{-13}$
GINE [5-fold bagged]	90/100	19/20	+7.53	+5.87	$p = 6.3 \times 10^{-13}$
D-MPNN (Chemprop)	89/100	19/20	+8.96	+6.42	$p = 8.7 \times 10^{-15}$
XGBoost (RDKit-2D)	91/100	20/20	+9.28	+8.44	$p = 6.5 \times 10^{-16}$
GINE	85/100	19/20	+9.35	+6.09	$p = 3.3 \times 10^{-11}$
XGBoost (ECFP4 + RDKit-2D)	90/100	20/20	+9.75	+7.36	$p = 1.7 \times 10^{-14}$
XGBoost (ECFP4 + RDKit-2D) [5-fold bagged]	89/100	20/20	+9.78	+6.82	$p = 7.8 \times 10^{-15}$
GCN	87/100	19/20	+10.66	+9.04	$p = 1.6 \times 10^{-12}$
GATv2	92/100	19/20	+11.40	+9.10	$p = 4.5 \times 10^{-15}$
RF (RDKit-2D)	91/100	20/20	+13.40	+13.59	$p = 5.3 \times 10^{-16}$
RF (RDKit-2D) [5-fold bagged]	93/100	20/20	+14.10	+13.68	$p = 5.2 \times 10^{-17}$
ChemBERTa-2 (77M-MTR) + linear	95/100	20/20	+20.77	+23.06	$p = 3.1 \times 10^{-17}$
RF (MACCS-167)	95/100	19/20	+21.46	+20.65	$p = 3.3 \times 10^{-16}$
ChemBERTa-2 (77M-MTR) + XGBoost	93/100	18/20	+23.41	+29.94	$p = 2.6 \times 10^{-16}$
XGBoost (ECFP4-2048)	96/100	20/20	+28.27	+32.56	$p = 1.0 \times 10^{-17}$
RF (ECFP4-2048)	93/100	20/20	+31.43	+38.16	$p = 1.5 \times 10^{-16}$

S6. Confirmation on fresh splits: all comparisons

Table S4: SIG-Net v4 vs. every router and cross-fitted expert on the confirmatory folds (seeds 1–3).

Comparator	Folds won	Mean (%)	Median (%)	p
Global stack (7) [primary]	35/57	+1.27	+0.20	$p = 0.020$
SIG-Net v3 router (7)	34/57	+0.91	+0.19	$p = 0.117$
Global stack (9)	34/57	+1.34	+0.21	$p = 0.019$
v4 w/o residual correction	29/57	+0.29	+0.01	$p = 0.016$
v4 w/o composite kernel	17/57	+0.38	+0.00	$p = 0.857$
v4 w/o novelty shrinkage	22/57	+0.02	+0.00	$p = 0.633$
D-MPNN + RDKit-2D [5-fold bagged]	48/57	+5.10	+2.43	$p = 7.3 \times 10^{-7}$
D-MPNN + Deep CORAL [5-fold bagged]	46/57	+5.54	+3.78	$p = 4.1 \times 10^{-7}$
GINE [5-fold bagged]	45/57	+5.70	+4.76	$p = 1.9 \times 10^{-5}$
AttentiveFP [5-fold bagged]	47/57	+5.09	+2.67	$p = 5.5 \times 10^{-7}$
XGBoost (ECFP4 + RDKit-2D) [5-fold bagged]	50/57	+10.10	+5.04	$p = 2.1 \times 10^{-9}$
RF (RDKit-2D) [5-fold bagged]	48/57	+13.91	+12.02	$p = 2.4 \times 10^{-8}$
Anchor (Multi-View Stack) [5-fold bagged]	42/57	+6.18	+3.06	$p = 1.2 \times 10^{-5}$
ChemBERTa-2 + XGBoost [5-fold bagged]	55/57	+22.87	+25.19	$p = 2.1 \times 10^{-10}$
XGBoost (ECFP4) [5-fold bagged]	54/57	+28.21	+32.41	$p = 2.0 \times 10^{-10}$
Original router vs Global stack (7)	28/57	+0.27	+0.00	$p = 0.805$

Global stack (7), FreeSolv generic included	38/60	+1.64	+0.26	$p = 0.0052$

S5. Per-setting results

Table S5: SIG-Net vs. the best other external model in each setting (mean ± SD over 5 seeds); paired t-test over seeds (descriptive; low power with $n = 5$).

Dataset	Split	SIG-Net	Best other	Model
BACE	BM scaffold	0.864 ± 0.046	0.856 ± 0.054	AttentiveFP [5-fold bagged]
BACE	Butina cluster	0.790 ± 0.076	0.800 ± 0.060	ChemBERTa-2 (77M-MTR) + XGBoost
BACE	Generic skeleton	0.815 ± 0.081	0.819 ± 0.048	ChemBERTa-2 (77M-MTR) + XGBoost
BACE	Random	0.900 ± 0.021	0.902 ± 0.021	Two-expert gate (single D-MPNN)
BBBP	BM scaffold	0.940 ± 0.029	0.940 ± 0.030	Two-expert novelty gate
BBBP	Butina cluster	0.887 ± 0.033	0.888 ± 0.024	Two-expert gate (single D-MPNN)
BBBP	Generic skeleton	0.954 ± 0.025	0.950 ± 0.017	XGBoost (ECFP4 + RDKit-2D) [5-fold bagge
BBBP	Random	0.940 ± 0.007	0.934 ± 0.011	Two-expert gate (single D-MPNN)
ESOL	BM scaffold	0.754 ± 0.112	0.746 ± 0.072	Two-expert gate (single D-MPNN)
ESOL	Butina cluster	0.544 ± 0.080	0.549 ± 0.084	Two-expert novelty gate
ESOL	Generic skeleton	0.704 ± 0.093	0.717 ± 0.096	Stack: XGBoost + D-MPNN
ESOL	Random	0.511 ± 0.042	0.515 ± 0.024	Stack: D-MPNN ens. + anchor
FreeSolv	BM scaffold	1.720 ± 0.136	1.714 ± 0.176	D-MPNN + Deep CORAL
FreeSolv	Butina cluster	1.038 ± 0.155	1.057 ± 0.133	Stack: XGBoost + D-MPNN
FreeSolv	Generic skeleton	1.405 ± 0.054	1.279 ± 0.058	GCN
FreeSolv	Random	0.800 ± 0.211	0.868 ± 0.168	Two-expert novelty gate
Lipophilicity	BM scaffold	0.580 ± 0.037	0.584 ± 0.039	AttentiveFP [5-fold bagged]
Lipophilicity	Butina cluster	0.668 ± 0.041	0.687 ± 0.057	AttentiveFP [5-fold bagged]
Lipophilicity	Generic skeleton	0.589 ± 0.033	0.605 ± 0.033	AttentiveFP [5-fold bagged]
Lipophilicity	Random	0.505 ± 0.047	0.527 ± 0.044	D-MPNN (Chemprop)